

$$FV_N = \text{CAD } 5,000,000$$

$$R_s = 6\% = 0.06$$

$$m = 12$$

$$R_s/m = 0.06/12 = 0.005$$

$$N = 10$$

$$mN = 12(10) = 120$$

$$\begin{aligned} PV &= FV_N \left(1 + \frac{R_s}{m}\right)^{-mN} \\ &= \text{CAD } 5,000,000 (1.005)^{-120} \\ &= \text{CAD } 5,000,000 (0.549633) \\ &= \text{CAD } 2,748,163.67 \end{aligned}$$

In applying Equation 8, we use the periodic rate (in this case, the monthly rate) and the appropriate number of periods with monthly compounding (in this case, 10 years of monthly compounding, or 120 months).

Annualizing Returns

To annualize any return for a period shorter than one year, the return for the period must be compounded by the number of periods in a year. A monthly return is compounded 12 times, a weekly return is compounded 52 times, and a quarterly return is compounded 4 times. Daily returns are normally compounded 365 times. For an uncommon number of days, we compound by the ratio of 365 to the number of days.

If the weekly return is 0.2 percent, then the compound annual return is 10.95 percent (there are 52 weeks in a year):

$$\begin{aligned} R_{\text{annual}} &= (1 + R_{\text{weekly}})^{52} - 1 = (1 + 0.2\%)^{52} - 1 \\ &= (1.002)^{52} - 1 = 0.1095 = 10.95\% \end{aligned}$$

If the return for 15 days is 0.4 percent, then the annualized return is 10.20 percent, assuming 365 days in a year:

$$\begin{aligned} R_{\text{annual}} &= (1 + R_{15})^{365/15} - 1 = (1 + 0.4\%)^{365/15} - 1 \\ &= (1.004)^{365/15} - 1 = 0.1020 = 10.20\% \end{aligned}$$

A general equation to annualize returns is given, where c is the number of periods in a year. For a quarter, $c = 4$ and for a month, $c = 12$:

$$R_{\text{annual}} = (1 + R_{\text{period}})^c - 1. \quad (8)$$

How can we annualize a return when the holding period return is more than one year? For example, how do we annualize an 18-month holding period return? Because one year contains two-thirds of 18-month periods, $c = 2/3$ in the above equation. For example, an 18-month return of 20 percent can be annualized as follows:

$$R_{\text{annual}} = (1 + R_{18\text{month}})^{2/3} - 1 = (1 + 0.20)^{2/3} - 1 = 0.1292 = 12.92\%.$$

Similar expressions can be constructed when quarterly or weekly returns are needed for comparison instead of annual returns. In such cases, c is equal to the number of holding periods in a quarter or in a week. For example, assume that you want to convert daily returns to weekly returns or annual returns to weekly returns for comparison between weekly returns. To convert daily returns to weekly returns, $c = 5$, assume that there are five trading days in a week. However, daily return calculations can be annualized differently. For example, five can be used for trading-day-based calculations, giving approximately 250 trading days a year; seven can be used on calendar-day-based calculations. Specific methods used conform to specific business practices, market